



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SREE AMMAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

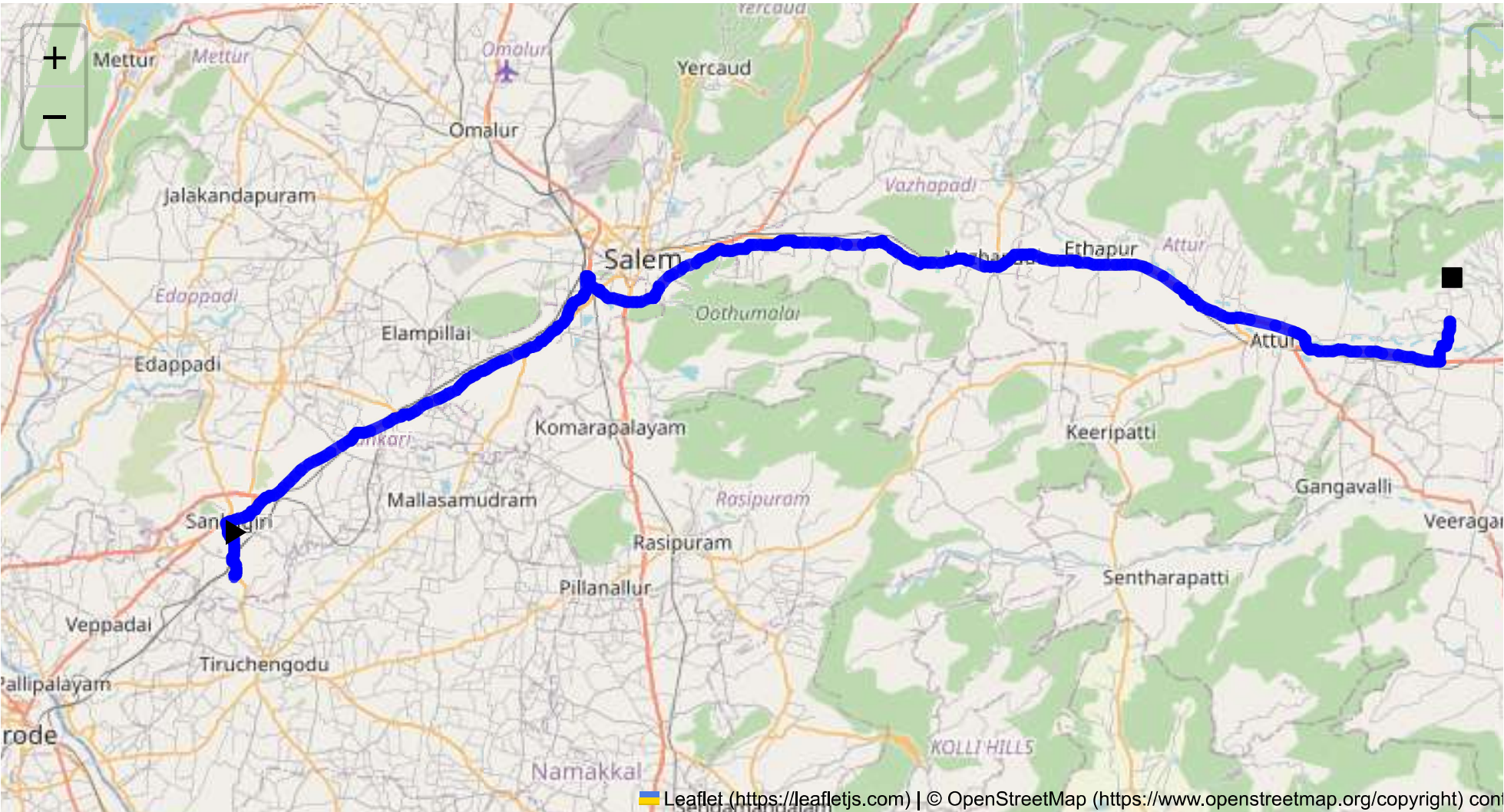
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 112.83 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours
Start: (11.4381, 77.8734)
End: (11.612525, 78.718451)



Welcome to the Journey Risk Management Study

Route Analysis Report

1. Overview of the Route Map:

- The route from CVQF+23W, Sangagiri to JP79+299, M.V Colony, spans approximately 112.83 kilometers.

- The journey typically starts southwards from Sangagiri, continues on NH 47, and transitions to NH 544 and NH 44 to reach the destination near M.V Colony.

2. Typical Weather Conditions and Potential Weather-related Hazards:

- Tamil Nadu experiences a tropical climate; hence common weather conditions include hot summers, mild winters, and seasonal monsoon rains (June to September).
- During the monsoon, the risk of slippery roads, reduced visibility, and flooding may increase.
- The region is generally humid with occasional cyclones that might affect road conditions.

3. Analysis of Traffic Patterns:

- Peak traffic hours typically align with commuting times: 8:00-10:00 AM and 5:00-8:00 PM.
- Congestion-prone areas on this route might include urban sections near Salem and Mallur.
- Highways usually see heavy vehicle traffic, especially around industrial areas and loading hubs.

4. Assessment of Road Quality and Infrastructure:

- Major highways (NH 544 and NH 44) are generally well-maintained, but certain stretches might suffer from potholes and uneven surfaces, especially following heavy rains.
- Pay close attention to signage regarding ongoing road work or detours.

5. Suggestions for Alternative Routes for Emergencies:

- A potential alternative is to divert through SH 86 at the point of major congestion.
- Hilly areas can also be avoided by taking routes that parallel NH47, ensuring better safety.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transporting hazardous materials requires adhering to Tamil Nadu's road safety regulations including specified lanes on highways, designated parking for breaks, and clearance to enter populated areas.
- Permit acquisition and adherence to permissible transport times are critical, especially near cities.

7. Overview of Historical Incidents:

- Historically, accidents involving heavy vehicles in the region often occur due to speeding and fatigue.
- Hazardous material incidents might involve spillage during monsoon due to slippery roads.

8. Environmental Considerations and Sensitive Areas:

- The route crosses ecologically sensitive areas including farmland and vicinity near water bodies.
- Hazmat transport should be particularly careful around these areas to prevent contamination and environmental disasters.

9. Analysis of Communication Coverage:

- Major highways on this route usually have robust communication coverage.
- Potential dead zones may exist in rural stretches between towns; ensuring alternative communication means (satellite phones or walkie-talkies) during these segments is advisable.

10. Estimated Emergency Response Times:

- Near urban areas like Salem, emergency response times may range from 15 to 30 minutes.
- In rural areas or during peak congestion, response times could extend up to an hour or more.

11. Overall Summary of Risk Assessment:

- This route presents moderate risks primarily due to traffic congestion, weather conditions, and specific segments of road maintenance issues.
- Hazmat transport should be particularly cautious during monsoon seasons and in congested urban areas.
- Proactive measures include real-time traffic monitoring, weather forecasts, securing proper permits, and having contingency plans for emergencies.
- Regular training on safety protocols, vehicle maintenance checks, and adherence to local transport regulations are recommended to mitigate potential risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.64172, 78.11817	30 KM/Hr
13	Turn	Medium	11.64216, 78.11875	30 KM/Hr
14	Turn	High	11.64195, 78.12129	15 KM/Hr
15	Turn	High	11.64120, 78.12168	15 KM/Hr
16	Turn	Medium	11.63948, 78.12081	30 KM/Hr
17	Turn	Medium	11.65703, 78.20465	30 KM/Hr
18	Blind Spot	Blind Spot	11.58480, 78.71206	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
19	Turn	High	11.59614, 78.71299	15 KM/Hr
20	Turn	High	11.59609, 78.71415	15 KM/Hr
21	Turn	High	11.59630, 78.71432	15 KM/Hr
22	Turn	High	11.59630, 78.71615	15 KM/Hr
23	Turn	High	11.59999, 78.71683	15 KM/Hr
24	Turn	High	11.60008, 78.71659	15 KM/Hr
25	Turn	High	11.60897, 78.71849	15 KM/Hr
26	Turn	High	11.60926, 78.71818	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
23	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
24	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
25	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
26	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
27	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
31	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
35	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
37	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
38	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
39	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
40	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
32	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
34	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
36	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium
41	school	Vani Vikas Matric Hr.sec school	11.6212713, 78.5462658	30 km/h	Medium
42	school	Saraswati Metric Hr sec School	11.5910827, 78.6402264	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

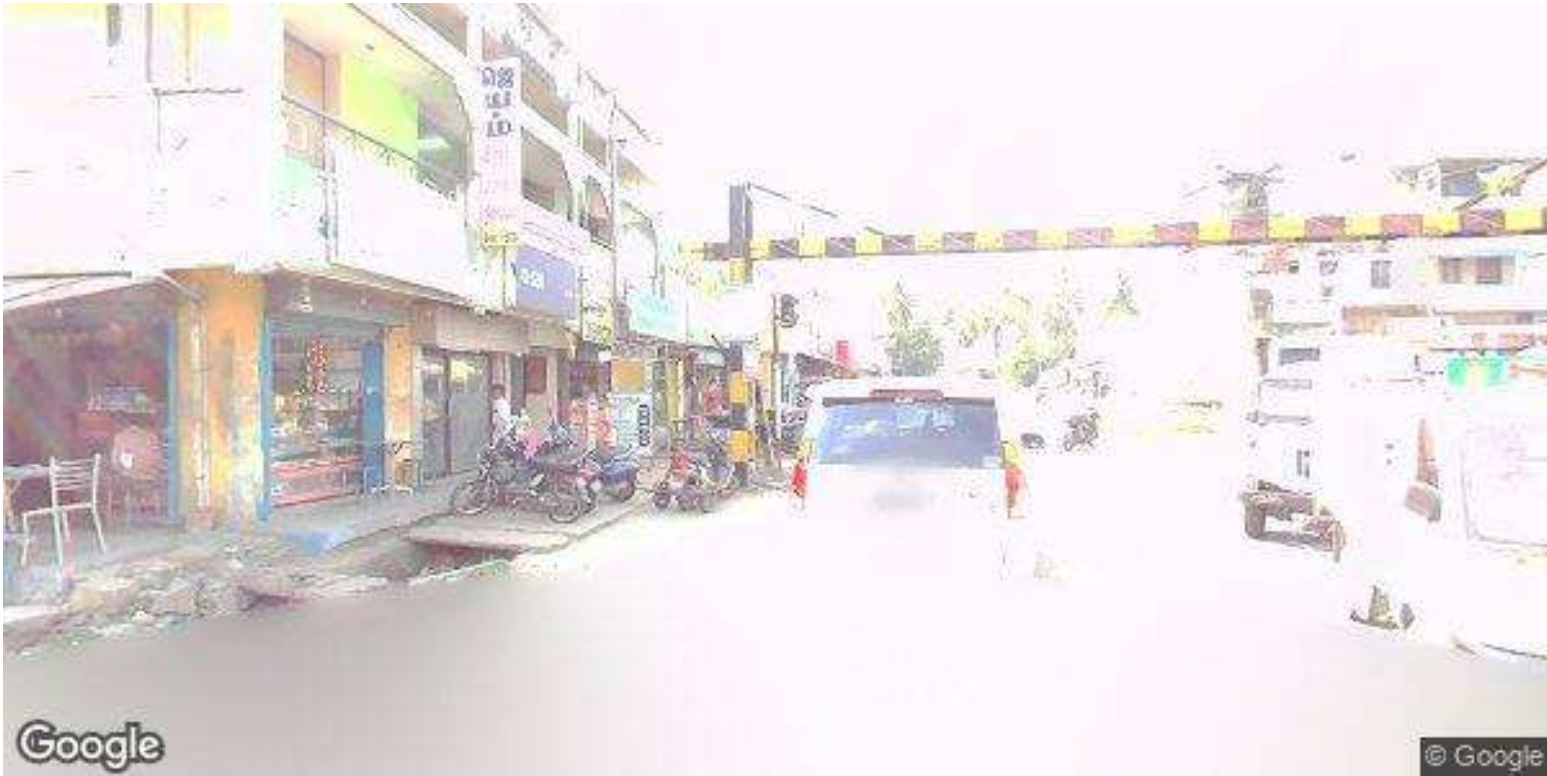


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65703, 78.20465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59614, 78.71299



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59609, 78.71415



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59630, 78.71432



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59630, 78.71615



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59999, 78.71683



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60008, 78.71659



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60897, 78.71849



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60926, 78.71818

